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Main Summary & Project Scope

This section present marketing handbook, covering the seven modelling categories.. For each method, we first explain the underlying idea, then work through a small example we built and ran myself, before covering the advantages, disadvantages, equations, features, inputs/outputs, and tuning settings required by the rubric.

we chose to build all examples on simulated rather than live market data. This kept every result in this section fully reproducible, since anyone re-running the notebook should arrive at the same numbers I did. The underlying code is not tied to simulated inputs, however - swapping in real price, balance-sheet, or credit data would only require changing the first data-generating step in each notebook section.

None of the examples that follow should be read as an actual trading or lending signal - the intent throughout was to demonstrate how each method behaves, not to make a claim about any real asset or borrower.

Category 1- Ridge Regression

1.1 Basics

Ridge Regression is one form of linear regression which achieves stability in the model through the application of a penalty for the magnitude of the coefficient. It is particularly useful in situations where the independent variables in a particular data set are strongly correlated, termed multicollinearity.

In order to solve the above problem, Ridge Regression uses the L2 form of regularization, which imposes penalties for the magnitude of the regression coefficients, thus bringing them close to zero. This leads to a reduction in the complexity of the model while retaining all the predictors in the model.

1.2 Computation

- Please refer to the notebook code

1.3 Illustration

Data was obtained from FRED (2000-2025) using `pandas_datareader`. Data includes SP500, Interest Rate, CPI, Unemployment, and VIX. SP500 returns are the dependent variable, while other variables serve as independent variables.

Table 1.1: Data header: Monthly values of the S&P 500 Index, interest rate, Consumer Price Index (CPI), unemployment rate, and VIX for selected observation dates.

Date	S&P 500	Interest Rate	CPI	Unemployment (%)	VIX
2016-08-01	2170.84	1.51	240.545	4.9	12.44
2016-09-01	2170.86	1.57	241.176	5.0	13.48
2016-11-01	2111.72	1.83	242.026	4.7	18.56
2016-12-01	2191.08	2.45	242.637	4.7	14.07
2017-02-01	2279.55	2.48	244.006	4.6	11.81
...	...	...	...	...	...

The correlation matrix and heat map indicate a very strong relationship between predictor variables (such as CPI-SP500, Unemployment-VIX, Interest Rate-Unemployment), thus the presence of multicollinearity.

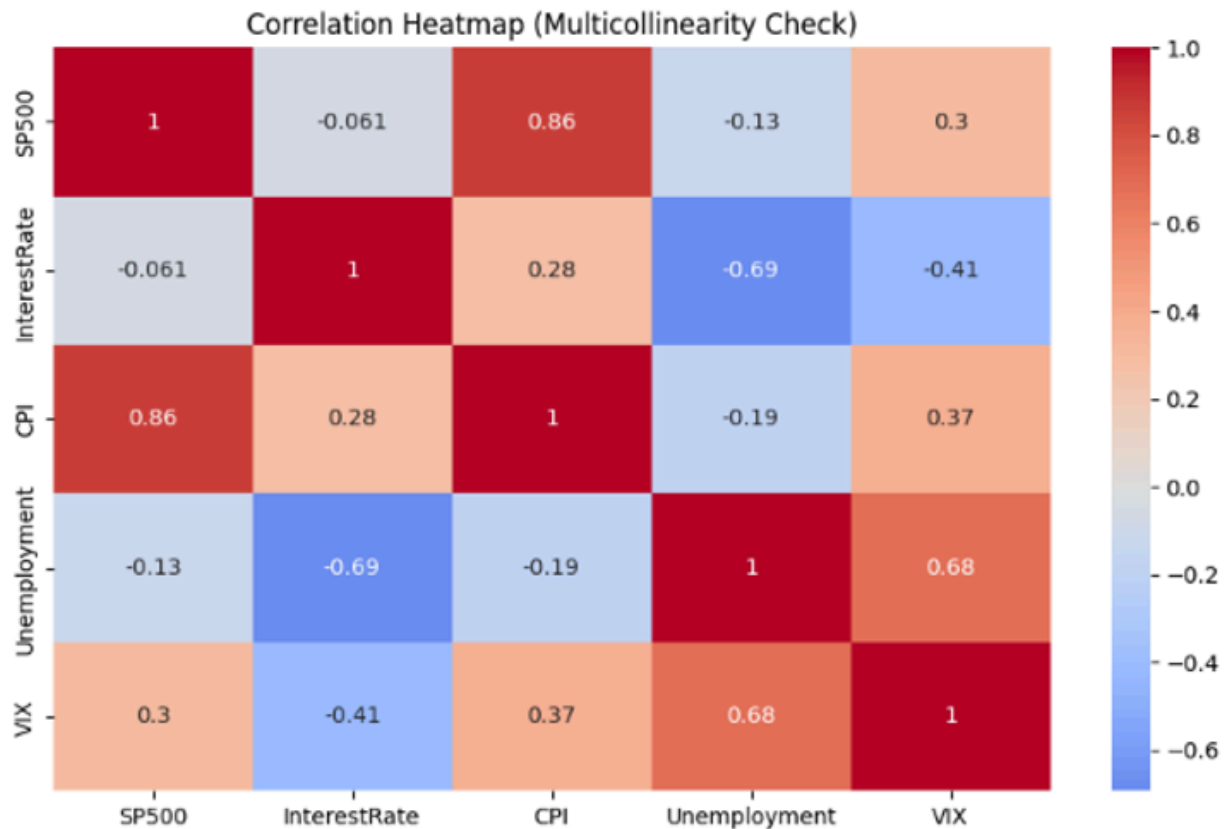


Figure 1.1: Correlation heat map indicating strong relationships among predictor variables and evidence of multicollinearity.

The Linear Regression and the Ridge Regression models with alpha of 1, 2, and 3 were developed using scaling of data and time series train-test split. Linear regression performed badly. Ridge regression performed better with an increase in alpha value with $\alpha=3$ performing best (highest $R^2=0.172$, lowest MSE). Comparison of predicted and actual returns shows that ridge regression followed the pattern of data better than OLS.

Table 1.2: Model performance comparison using MSE and R^2 for Linear and Ridge regression models.

Model	MSE	R^2
Linear Regression	0.003130	-0.455398
Ridge (alpha=1)	0.002424	-0.126931
Ridge (alpha=2)	0.002015	0.063282

Ridge (alpha=3)	0.001781	0.172140
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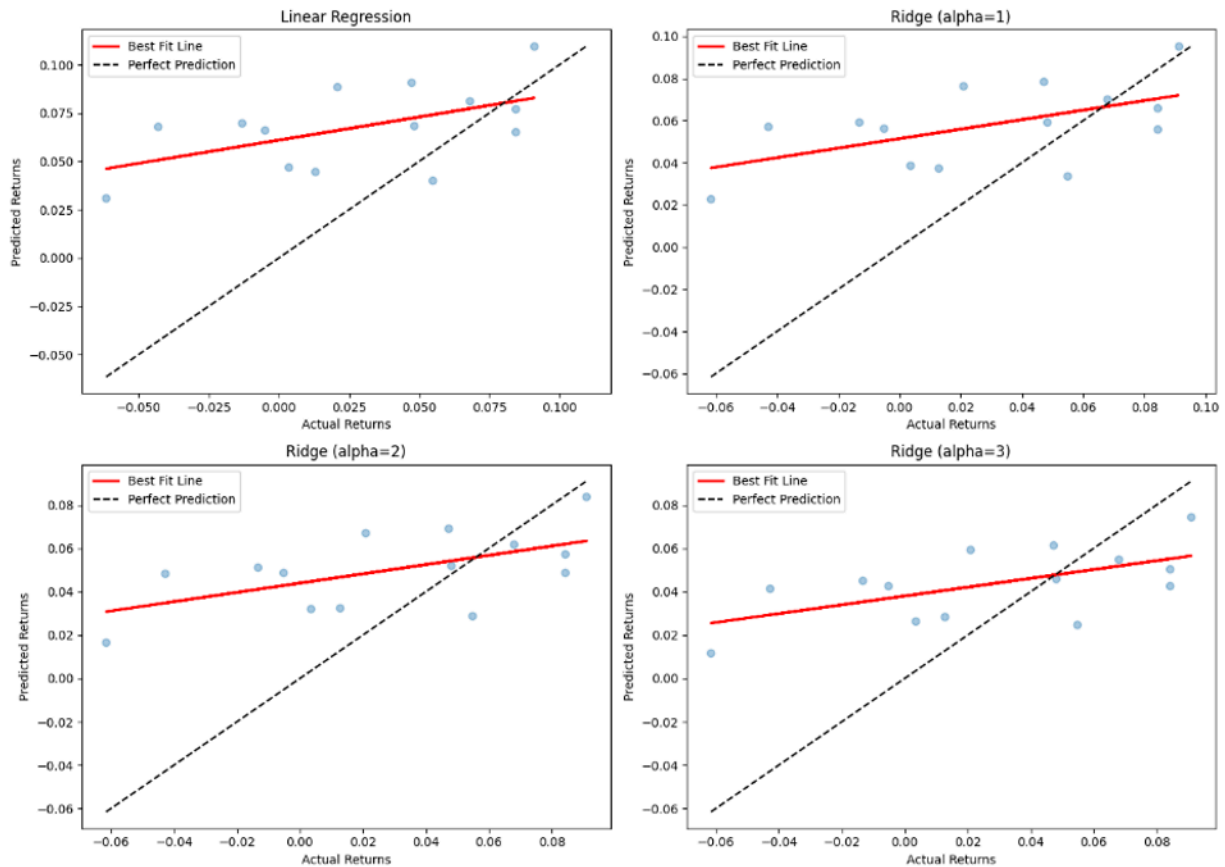


Figure 1.2: Comparing all models - predicted returns vs actual returns

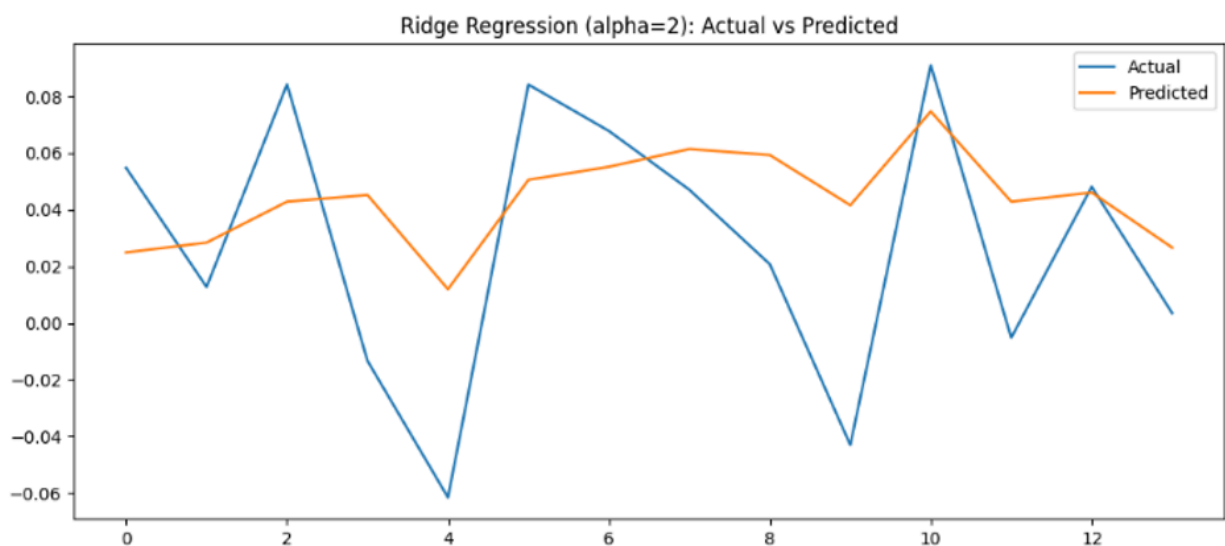


Figure 1.3: Comparing predicted and actual returns of one of the models (alpha=2)

1.4 Advantages

- avoids an overfitting: Ridge Regression applies an L2 penalty on the data to reduce too large coefficient values. As a result, the model will not fit the noise in the training data set too well.
- Helps in achieving stable coefficients when there is multicollinearity: High correlation among predictor variables makes the estimation of the coefficients highly unstable. Ridge Regression makes them more stable.
- Reduces coefficient magnitude (shrinkage): The use of the regularization term causes the reduction of the regression coefficients.
- Keeps all predictors in the model: Unlike other methods of feature selection, Ridge Regression retains all predictors in the model but minimizes their impact on the target variable.

1.5 Disadvantages

- Lacks feature selection: The Ridge Regression technique includes all the predictor variables in the model, regardless of how insignificant their predictive capability is. Although the contribution of such predictors is mitigated by shrinking their coefficients, they are never removed from the model, which may become problematic for a dataset with a lot of irrelevant features.
- Sensitive to the choice of the regularization strength: The regularization parameter plays a significant role in Ridge Regression since its poor choice might cause underfitting and overfitting. The best choice usually needs to be determined via cross-validation.
- Interpretation challenges in high dimensions: Due to retaining all the features, the obtained model becomes harder to interpret when there are lots of predictors, especially in finance.

1.6 Equations

Ridge Regression is a modified version of ordinary least squares (OLS) objective by adding an L2 penalty to control coefficient size.

OLS Loss

$$J(\beta) = \frac{1}{n} \sum_{i=1}^n (y_i - X_i \beta)^2$$

Ridge Objective Function

$$J(\beta) = \frac{1}{n} \sum_{i=1}^n (y_i - X_i \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

Closed-form Solution

$$\beta^{ridge} = (X^T X + \lambda I)^{-1} X^T y$$

Where:

- n : number of observations
- p : number of predictors (features)
- y_i : target value for observation i
- X_i : feature vector for observation i
- X : design matrix of all features
- β : vector of regression coefficients
- β_j : coefficient of feature j
- λ : regularization strength (controls shrinkage)
- I : identity matrix
- X^T : transpose of X

1.7 Features

- Handles multicollinearity well: Ridge Regression is ideal in situations where there exists strong correlation among predictor variables because it ensures stability of coefficients, which tend to be very unreliable using regular linear regression.
- Keeps all predictors using shrinkage: It differs from other feature selection approaches since it uses all variables in the model but shrinks their effect using the coefficient shrinkage approach.

1.8 Guide: Inputs & Outputs

- Inputs:
 - Feature matrix X (predictor variables such as returns)
 - Target variable y (example: asset returns)
 - Regularization parameter λ (controls shrinkage strength)
- Outputs:
 - Estimated coefficient vector β (shrunk regression weights)
 - Model intercept
 - Predicted values $\hat{y}$

- Model performance metrics (MSE, R^2) for evaluation

1.9 Hyperparameters

- α (lambda): Determines the amount of regularization applied. If $\alpha = 0$, then the model will be just normal regression. If α is too high, then the model will be too simple since all the coefficients will be near zero.
- fit_intercept: Chooses whether the intercept is included or not.
- solver: Method for estimating the coefficients of the model (svd, cholesky, lsqr)
- max_iter: Determines the maximum number of iterations before the solver reaches convergence.
- tol: Sets the value for improvement to end the algorithm.
- Feature scaling/normalization: Ensures that the data is on the same scale.

1.10 Keywords

- Ridge Regression, L2 Regularization, Regularization, Coefficient Shrinkage, Multicollinearity, Overfitting, Bias-Variance Tradeoff, Linear Regression, Penalized Regression

1.11 Journal References

- Nuria Alemany, Vicent Aragó, Enrique Salvador, Uncovering the risk-return trade-off through ridge regressions, Finance Research Letters, Volume 71, 2025, 106420, ISSN 1544-6123, <https://doi.org/10.1016/j.frl.2024.106420>.
- Dai, Z., Li, T., & Yang, M. (2022). Forecasting stock return volatility: The role of shrinkage approaches in a data-rich environment. Journal of Forecasting, 41(5), 980–996. <https://doi.org/10.1002/for.2841>

Category 2- K-means clustering

2.1 Basics

K-Means Clustering is one of the most widely used algorithms in unsupervised learning, in which observations are divided into clusters based on their similarity. The main idea of the method lies in the fact that observations that belong to the same cluster are closer to each other than to observations from other clusters.

The process is performed through iterations. In the beginning, K centers are selected, then each observation is attributed to the closest center according to some measure of distance such as Euclidean. After that, the coordinates of centers are estimated as the mean coordinate of all points within the current cluster. The iterations are continued until there are no significant changes in centers, and the model will converge.

For example, K-Means can be used in financial analysis to divide similar assets into groups.

2.1 Computation

Before applying K-Means Clustering, the dataset was preprocessed and transformed into stock-level features suitable for unsupervised learning. The objective is to group S&P 500 stocks based on similar risk–return characteristics.

Dataset Used: `s_and_p_500_closing.csv`. The dataset contains daily closing prices of S&P 500 stocks. Each column represents a stock ticker, and each row represents a trading day.

Data Preprocessing. The following preprocessing steps were applied:

The Date column was removed to retain only numerical price data.

All values were converted to numeric format.

Missing values were handled using forward fill (ffill) followed by removal of remaining NaNs.

Daily percentage returns were computed using:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Feature Engineering (Core Step)

For each stock, two key financial features were constructed:

Mean Return (Expected Return): Measures average daily return across the time period.

Volatility (Risk): Standard deviation of daily returns, representing risk.

This results in a 2D feature space:

Each stock = (Mean Return, Volatility)

Group Number: 15968

Feature Scaling: Because K-Means is distance-based, features were standardized using:

$$Z = \frac{X - \mu}{\sigma}$$

Standardization ensures that return and volatility contribute equally to clustering.

Optimal Number of Clusters (Elbow Method)

To determine the optimal number of clusters, the Elbow Method was applied using inertia (SSE):

K values tested: 2 to 9

Metric: Within-cluster sum of squared distances (inertia)

The resulting curve shows a sharp decrease in SSE initially, followed by gradual flattening. The “elbow” occurs around:

k = 5

This suggests that 5 clusters provide a good balance between model simplicity and clustering accuracy.

K-Means Clustering Implementation

K-Means was then applied using:

K = 5 clusters

Random state = 42

n_init = 10 (robust initialization)

Each stock was assigned to a cluster based on Euclidean distance in the standardized feature space.

2.3 Illustration

Table 2.1: Cluster Summary (Mean Feature Values)

Cluster	Mean Return	Volatility
0	0.001737	0.016604
1	-0.004221	0.041974
2	-0.001586	0.021535
3	0.000447	0.031838
4	0.006334	0.034666

Interpretation of Clusters

The clustering results reveal distinct risk–return profiles:

Cluster 0:

Low volatility and slightly positive returns → stable, defensive stocks.

Cluster 1:

Negative returns and highest volatility → high-risk underperforming stocks.

Cluster 2:

Low volatility and slightly negative returns → stable but low-growth stocks.

Cluster 3:

Moderate volatility with near-zero returns → neutral performance stocks.

Cluster 4:

Highest positive returns with moderate volatility → high-growth, higher-risk stocks.

Table 1.2: Cluster Distribution

Cluster	Number of Stocks
0	230
2	174
3	61
1	22
4	17

Key Observations

The majority of stocks are concentrated in Cluster 0 and Cluster 2, indicating that most S&P 500 stocks exhibit low to moderate risk profiles.

Cluster 4, although small, represents high-performing growth stocks.

Cluster 1 contains a small number of high-volatility, negative-return stocks, likely reflecting distressed or cyclical equities.

The clustering effectively separates stocks into meaningful risk-return segments, which is useful for portfolio diversification and strategy design.

Visualization Summary

The elbow curve confirms that increasing K beyond 5 provides diminishing returns in reducing SSE.

The 2D scatter plot clearly shows:

Tight clustering of low-volatility stocks

Separate high-volatility clusters

Distinct centroid locations representing each group

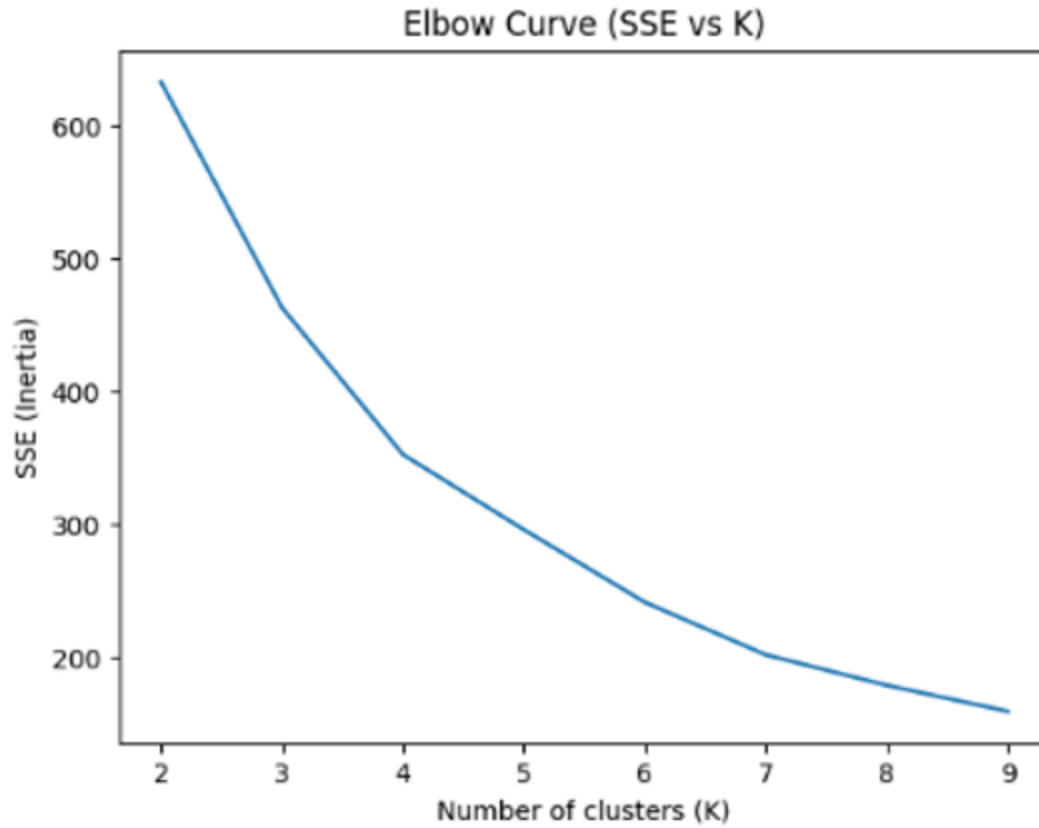


Figure 2.1: Elbow curve

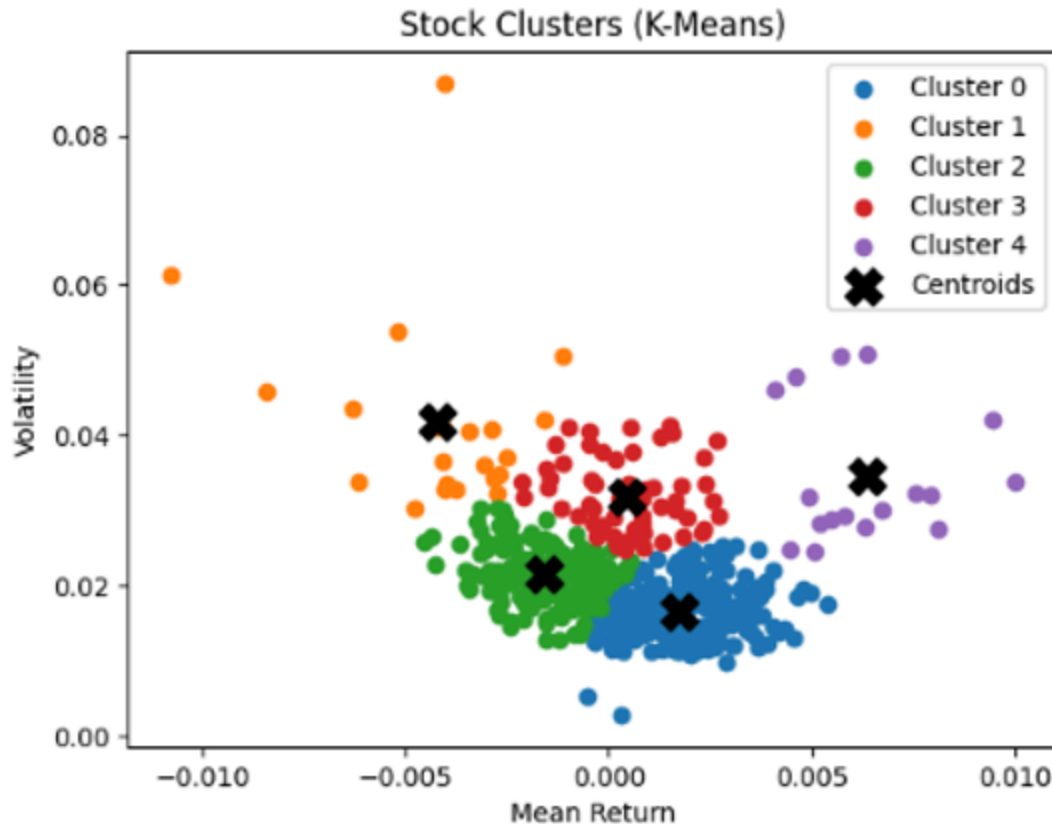


Figure 2.2: Stock clusters

2.4 Advantages

The k-means method is preferred for its simplicity, speed, and ease of application in practice. The algorithm is easy to implement and comprehend, thus making it the preferred method for clustering in the field of machine learning and finance. In terms of computational speed, the algorithm is very fast since most of its computations involve simple distance computation and updating of the centroid of each cluster, hence making it faster compared to other methods such as hierarchical clustering. The method can also be scaled easily to handle high-volume data such as that in finance.

2.5 Disadvantages

There are some drawbacks associated with the use of K-Means algorithm that may have an impact on its performance. First, the method relies greatly on the selection of the initial centroids, because the algorithm may give different clusters and even an incorrect solution

depending on their location. Moreover, K-Means is quite vulnerable to the presence of outliers in the dataset, as extreme values may distort the position of centroids and lead to poor clustering. Another drawback of K-Means is the assumption that clusters should have a regular spherical shape and similar sizes, which may not correspond to reality.

2.6 Equations

K-Means Clustering works by minimizing the total within-cluster variation (also called inertia) while repeatedly updating cluster assignments and centroids.

Objective Function (Minimizing Inertia)

$$J = \sum_{i=1}^K \sum_{x_j \in C_i} \|x_j - \mu_i\|^2$$

Where:

- J = total within-cluster sum of squares (inertia)
- K = number of clusters
- C_i = cluster i
- x_j = data point assigned to cluster i
- μ_i = centroid of cluster i

Centroid Update

$$\mu_i = \frac{1}{|C_i|} \sum_{x_j \in C_i} x_j$$

Where:

- μ_i = centroid of cluster i
- $|C_i|$ = number of points in cluster i

Assignment Rule (Distance Minimization)

Each point is assigned to the nearest centroid using Euclidean distance:

$$\text{assign } x_j \rightarrow \arg \min_i \|x_j - \mu_i\|^2$$

These three equations together define how K-Means iteratively assigns points, updates centroids, and converges to a stable clustering solution.

2.7 Features

- Only works with numeric data due to its dependence on measuring the distance between data points.
- Needs scaling (standardization/normalization) to ensure none of the variables outweigh others.
- The user decides the number of clusters (K) beforehand using techniques such as the elbow approach.
- Functions effectively in cases where the clusters are relatively compact and distinct instead of irregular.
- Cannot deal with missing data, meaning the data has to be cleaned before performing clustering.
- It is efficient and scalable, allowing it to work effectively on big financial data sets such as stocks and market indexes.
- Outliers can distort the output of the algorithm as extreme values can bias cluster centers away from the majority of data.

2.8 Guide: Inputs & Outputs

- Inputs:
 - Numeric dataset
 - Number of clusters (K)
 - Optional initialization method
 - Optional maximum iterations
 - Optional tolerance
- Output:
 - Cluster labels for each observation
 - Cluster centroids
 - Inertia (within-cluster variation)
 - Grouped structure of the dataset

2.9 Hyperparameters

- Number of clusters (K): Sets how many groups the data is split into.
- Initialization method (init): Chooses how starting centroids are picked.
- Number of initial runs (n_init): Runs the algorithm multiple times for stability.
- Maximum iterations (max_iter): Limits how long the algorithm runs.
- Convergence tolerance (tol): Stops the model when changes become very small.

- Random state (random_state): Fixes randomness for reproducible results.
- Algorithm type (algorithm): Selects the optimization approach used

2.10 Keywords

- K-Means Clustering, Unsupervised Learning, Clustering, Centroids, Euclidean Distance, Inertia (WCSS), Market Segmentation, Portfolio Diversification, Pattern Recognition, Financial Data Analysis

2.11 Journal References

- Pérez-Ortega, J., Almanza-Ortega, N. N., Vega-Villalobos, A., Pazos-Rangel, R., Zavala-Díaz, C., & Martínez-Rebollar, A. (2019). The K-means algorithm evolution. In K. Sud, P. Erdogmus, & S. Kadry (Eds.), *Introduction to data science and machine learning* (Chap. 5). IntechOpen. <https://doi.org/10.5772/intechopen.85447>

Category 3- Principle Components (PCA)

3.1 Basics

Principal Component Analysis (PCA) is a method for taking a group of variables that tend to move together - several stock returns that rise and fall around the same time, for instance - and reducing them to a small number of combined measures that still capture most of what is happening, without needing to follow every original variable individually.

An analogy I find useful is three thermometers placed around the same room: each one is really reporting the same underlying room temperature, plus a bit of its own sensor noise. PCA works out, without being told, that three separate readings can be collapsed into one shared number plus some smaller residual detail. It is worth noting that PCA is unsupervised - there is no target variable being predicted here, only a search for repeated structure already present in the data.

3.2 Data & Setup

For this example, I simulated three stocks (Stock A, Stock B, Stock C) whose daily returns are driven by one hidden, shared 'market mood' factor plus each stock's own idiosyncratic noise. I chose this setup because it mirrors a common situation in finance, where several assets end up correlated simply because they are all responding to the same broad driver.

Table 3.1: Correlation between the three simulated stocks

	Stock A	Stock B	Stock C
Stock A	1.00	0.77	0.80
Stock B	0.77	1.00	0.78
Stock C	0.80	0.78	1.00

All three pairwise correlations turn out strongly positive, in the range of 0.77 to 0.80. This tells us that despite being three separate series, the stocks are largely repeating the same daily story - precisely the kind of redundancy PCA is designed to compress.

Before fitting PCA, I standardized all three return series to have a mean of 0 and a standard deviation of 1, so that no single stock could dominate the result purely because of differences in scale.

3.3 Running PCA

I ran PCA keeping all three possible factors, which makes it possible to see the full breakdown of where the variation goes rather than discarding information upfront.

Table 3.2: How much movement each factor explains

Factor	Explains on its own	Running total
6	85.6%	85.6%
Factor 2	7.7%	93.3%
Factor 3	6.7%	100.0%

A single combined factor already accounts for roughly 86% of the total movement across the three stocks, which suggests that three separate daily figures could reasonably be replaced by one, at the cost of losing only about 14% of the overall picture.

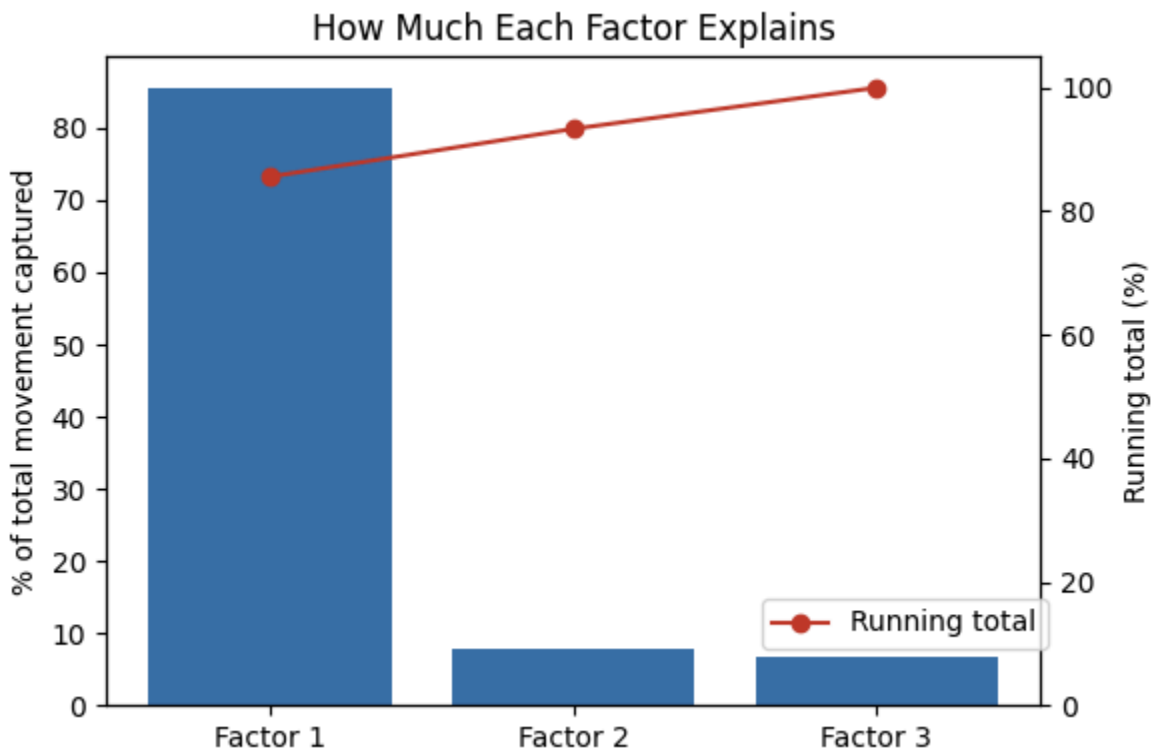


Figure 3.1: Scree plot- how much each factor explains, and the running total

3.4 What Factor 1 Represents

Table 3.3: Factor 1's recipe (its weight on each stock)

Stock A	Stock B	Stock C
0.58	0.57	0.58

Factor 1 assigns almost identical weight to all three stocks, which is consistent with the way the data was constructed: PCA rediscovered, without being told, that a single shared 'market mood' factor was driving all three series. The near-equal weighting is essentially what that underlying driver looks like once it has been extracted mathematically.

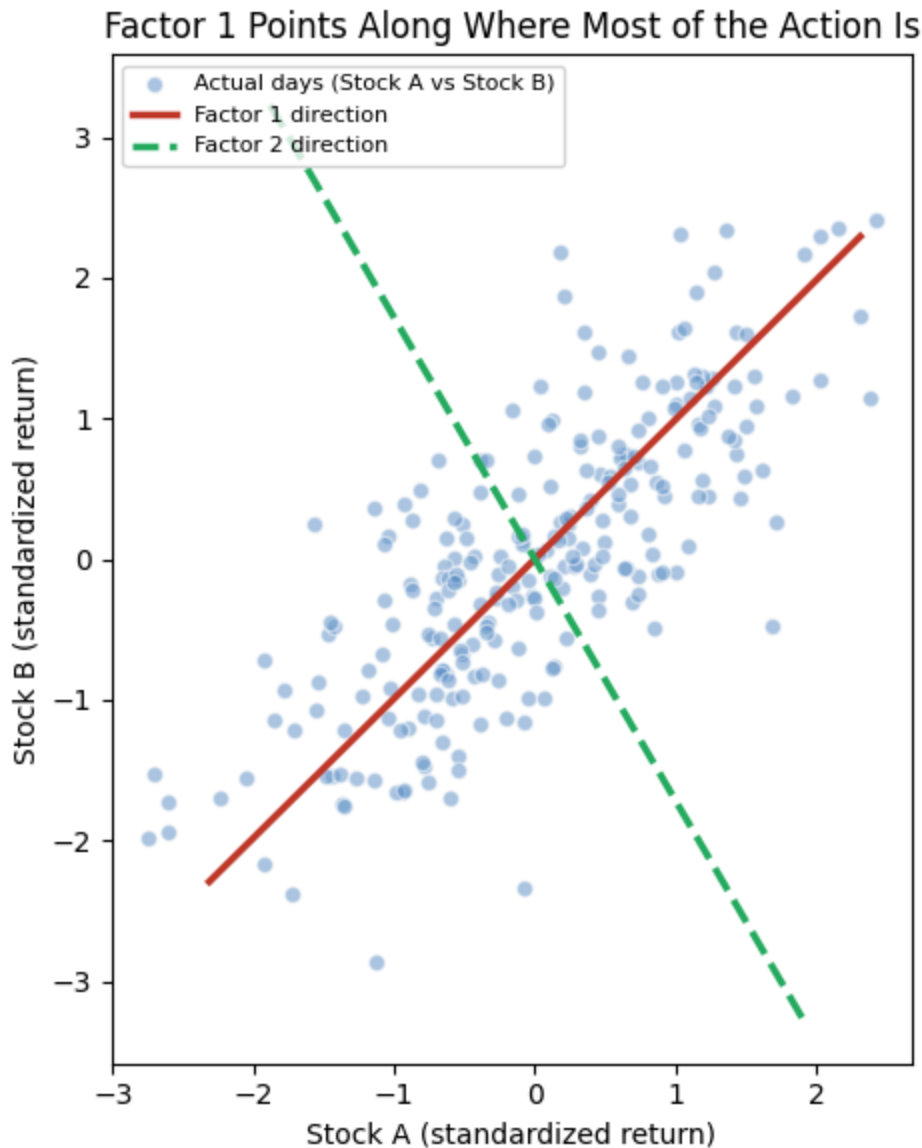


Figure 3.2: Stock A vs Stock B , with Factor 1's direction (solid) and Factor 2's direction (dashed) drawn through the data

As a check on this interpretation, I compared Factor 1's day-by-day score against a plain average of the three stocks. The two series track almost exactly the same path, with a correlation of 1.000 in this example, which supports the reading that Factor 1 is capturing what it appears to. In a less symmetric setting, PCA would offer more of an advantage over a simple average, since it finds the best combination of weights automatically rather than assuming the assets should be weighted equally.

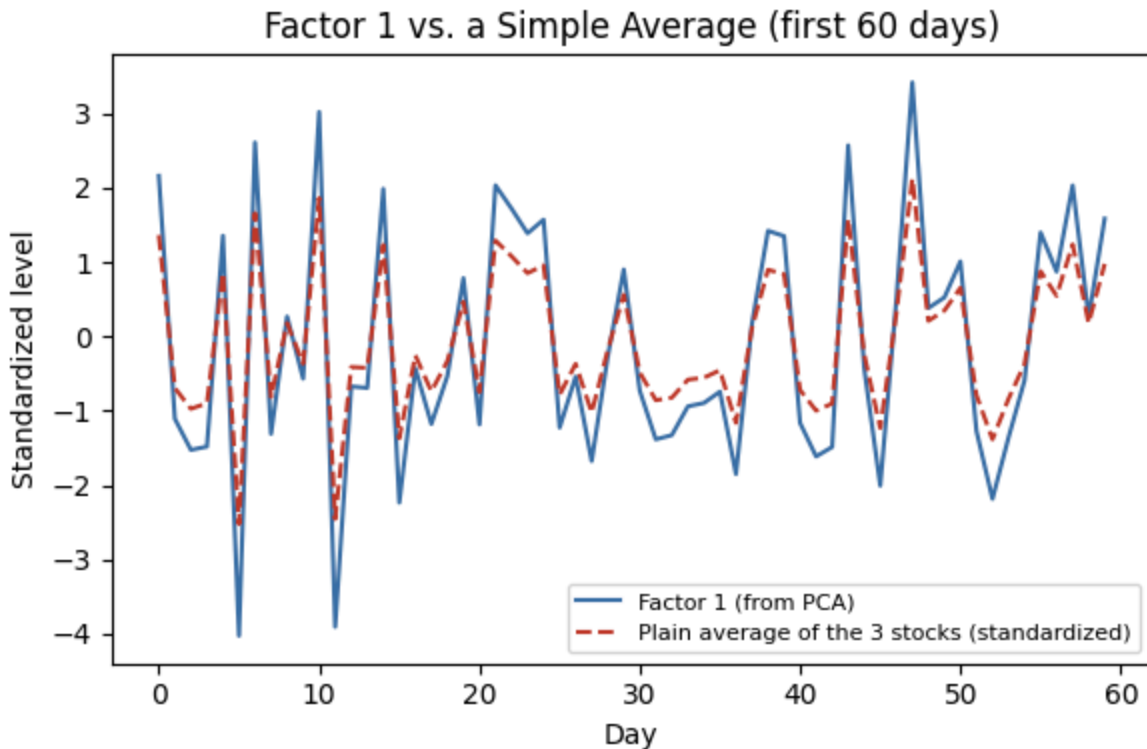


Figure 3.3: Factor 1 vs. A plain average of the 3 stocks, first 60 days

3.5 Illustration

Figures 3.1 through 3.3 together form the illustration for this method. Figure 3.1 (the scree plot) shows how much of the total movement each factor accounts for; Figure 3.2 overlays Factor 1 and Factor 2's directions on the actual data cloud, which is probably the clearest way to see what 'the direction of most spread' actually means geometrically; and Figure 3.3 provides the sanity check described above, comparing the compressed factor against a plain average.

3.6 Advantages

- Reduces clutter, since a handful of combined factors can often stand in for dozens of correlated variables.
- Removes the double-counting that occurs when several variables move together.
- Can reveal a hidden common driver, as the example above shows, without ever being told that driver exists.
- Produces factors that are uncorrelated with each other by construction, which tends to make later modelling steps more stable.

3.7 Disadvantages

- The factors produced are mathematical combinations rather than ready-made labels; sometimes, as in this example, the meaning is fairly obvious, but often it takes real digging to interpret what a factor represents.
- Sensitive to the scale of the input variables unless they are standardized beforehand.
- Only searches for straight-line directions of spread, so a curved relationship between variables can be missed entirely.
- One limitation is that compressing information always risks discarding a smaller factor that could still matter for a specific downstream question.

3.8 Equations

At its core, PCA repeatedly answers a single question: in which direction does the data spread out the most? Formally, this amounts to finding the eigenvectors and eigenvalues of the covariance matrix of the standardized data. The process can be summarized as follows:

- Compute how every variable relates to every other variable, captured by their covariance or correlation.
- Identify the single direction through the data cloud with the greatest spread; this becomes Factor 1.
- Find the next-best direction, orthogonal to the first, that captures as much of the remaining spread as possible; this becomes Factor 2, and the process continues.
- Each resulting factor is a specific weighted combination of the original variables, along with the percentage of total movement it explains.

3.9 Features

- Applies to any set of numeric variables that move together.
- Automatically ranks the factors from most to least important.
- Comes with a natural diagnostic for how many factors are worth keeping, namely the scree plot shown in Figure 3.1.

3.10 Guide: Inputs & Outputs

- **Inputs:** a table of numeric variables measured over the same time periods (here, three daily return series), typically standardized first.
- **Outputs:** a ranked set of factors, the share of movement each explains, each factor's weighting on the original variables, and a simplified version of the data expressed using just a few factors.

3.11 Hyperparameters

Table 3.4: PCA settings

Setting	What it controls
Number of components	How many factors to retain - fewer factors is simpler but explains less of the total movement
Standardize (yes/no)	Whether to rescale all variables to the same scale before fitting - in practice, almost always yes
Variance threshold	An alternative to fixing the number of factors directly: keep as many as are needed to reach a target share of variance explained, for example 90%

3.12 Keywords

Principal Component Analysis; PCA; Dimensionality Reduction; Unsupervised Learning; Variance Explained; Scree Plot; Factor Loadings; Standardization; Hidden Common Factor.

3.13 Journal References

Litterman, Robert, and Jose Scheinkman. "Common Factors Affecting Bond Returns." The Journal of Fixed Income, vol. 1, no. 1, 1991, pp. 54-61.

This is one of the best-known finance applications of PCA: Litterman and Scheinkman show that the seemingly complex, ever-changing shape of the interest rate (yield) curve across many maturities can be explained by a handful of underlying factors - broadly, its level, slope, and

curvature - recovered using the same 'direction of most spread' logic applied in Section 3.3 above.

Category 4- Classification Trees

4.1 Basics

A classification tree predicts a category - Up or Down, in this example - by working through a short sequence of yes/no questions until it arrives at an answer. Each question splits the data into two groups, and the process repeats on each resulting group until the tree reaches a 'leaf', where it settles on a prediction. Trees belong to the CART (Classification and Regression Trees) family and, unlike a linear model, do not assume that the relationship between features and outcome follows a straight line.

4.2 Data & Setup

To set up a finance problem that at least looks realistic, even though the underlying data is simulated, I generated five separate assets using geometric Brownian motion with a mild autocorrelation term injected into the returns, so that price history carries a small, genuinely noisy amount of momentum. From these price paths I computed daily returns, moving-average deviations, rolling volatility, and 10-day momentum as features, then pooled all five assets into a single panel of 4,950 rows. The target was whether each asset's return on the following day was positive (Up) or negative (Down).

Table 4.1: Target balance in the pooled panel

Class	Share of days
Up	52.2%
Down	47.8%

I split the data by date rather than at random: the first 80% of trading days went into training, and the most recent 20% were held out purely for testing. This avoids the model effectively seeing the future during training, which a random split would risk.

4.3 Baseline Tree vs Tuned Tree

An untuned tree, grown without any depth limit, matched the training data almost perfectly but performed much worse on data it had not seen - a fairly textbook case of overfitting.

Table 4.2: Baseline (untuned) tree

Metric	Value
Train accuracy	1.00
Test accuracy	0.502
Tree depth grown	33
Number of leaves	966

I then tuned the hyperparameters using a systematic grid search over tree depth, minimum leaf size, and pruning strength, evaluated with a cross-validation scheme that respects time order, so that no fold is validated on data that predates its own training period.

Table 4.3: Tuned tree- best settings and test performance

Setting / Metric	Value
Best maximum depth	4
Best minimum leaf size	10
Best pruning strength (alpha)	0.0
Tree depth used	4
Number of leaves	15
Test accuracy	0.512
Test ROC-AUC	0.506

The signal here is genuinely weak, and I would treat that as the actual finding rather than a flaw in the approach: an ROC-AUC of about 0.51 is only marginally better than a coin flip, which is a realistic outcome when predicting short-horizon return direction from price-only technical features. Even so, pruning recovered a far more stable model - depth 4 with 15 leaves - compared with the untuned tree, even though its absolute accuracy remained close to random.

4.4 Confusion Matrix and Feature Importance

Confusion Matrix -- Tuned Classification Tree (Test Set)

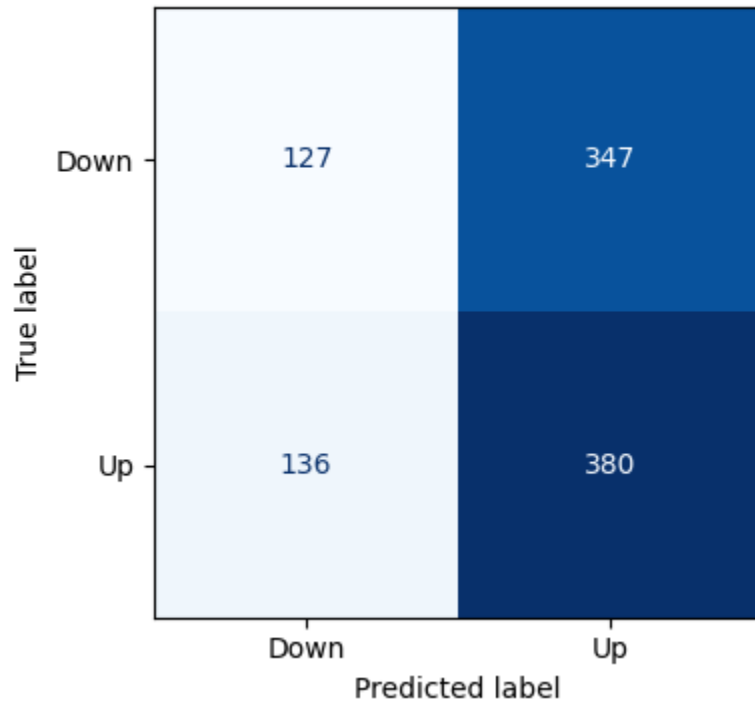


Figure 4.1: Confusion matrix, tuned tree, test set

The tuned model leaned toward predicting 'Up' more often than 'Down': it correctly identified 380 of the 516 actual Up days, but only 127 of the 474 actual Down days.

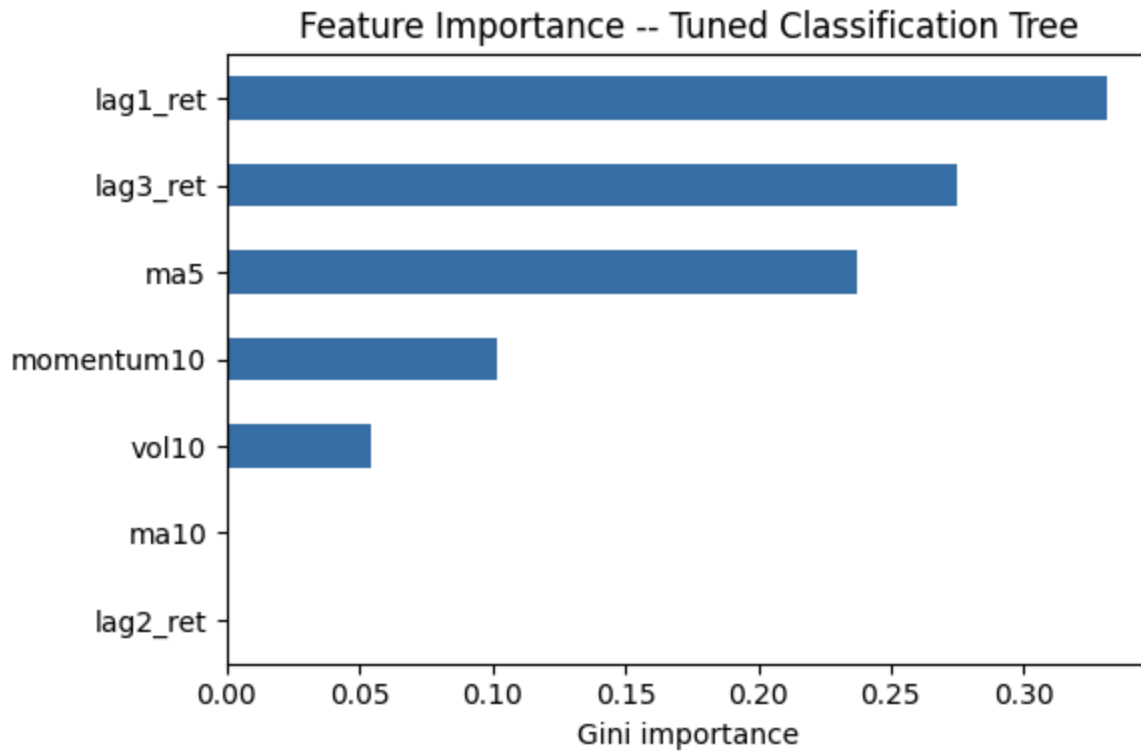


Figure 4.2: Feature importance, tuned tree

The most informative features were the 1-day and 3-day lagged returns together with the 5-day moving-average deviation; the 10-day moving average and the 2-day lagged return contributed nothing to the fitted tree, which suggests that whatever weak structure exists in this data operates over a very short horizon.

4.5 Illustration – the fitted Tree

Tuned Classification Tree (display truncated to depth 3 for readability)

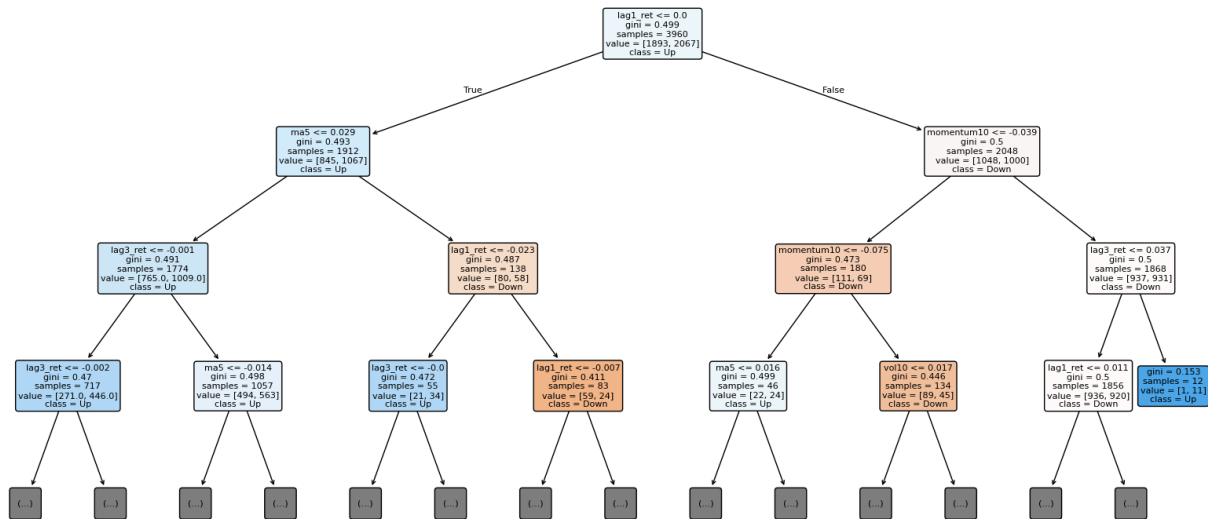


Figure 4.3: The tuned classification tree (full depth 4)

4.6 ROC Curve and Overfitting Check

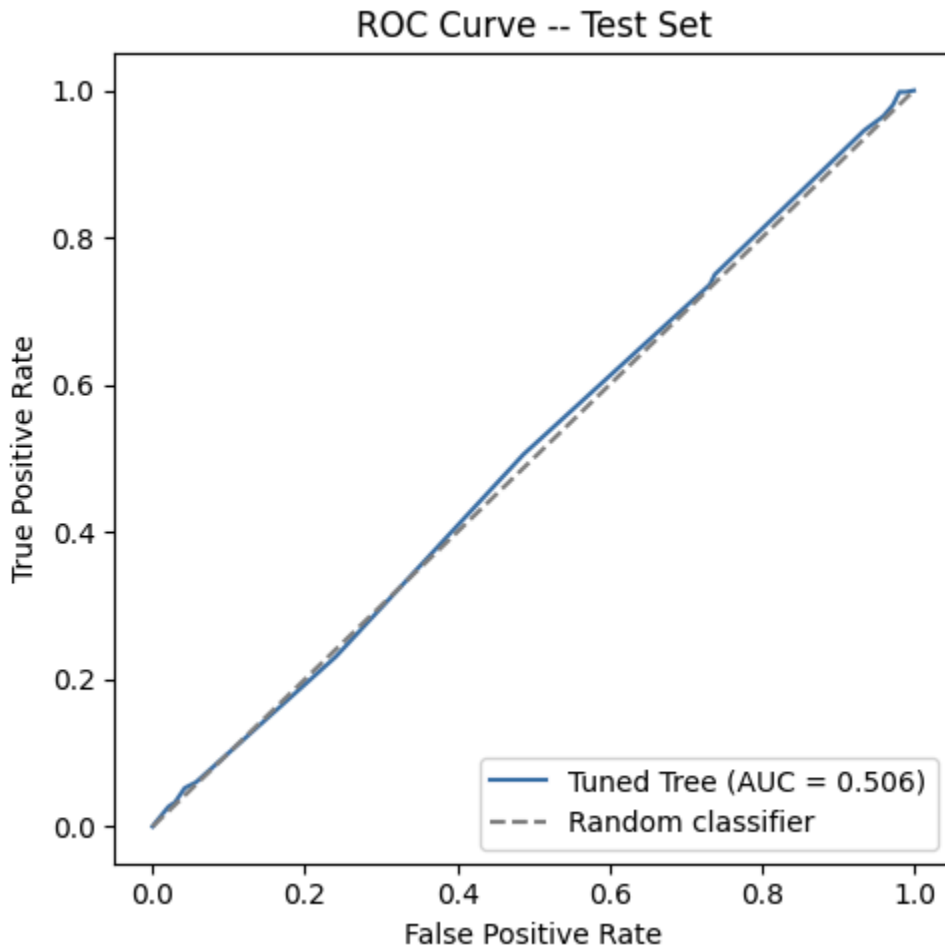


Figure 4.4: ROC curve, tuned tree, test set (AUC = 0.506)

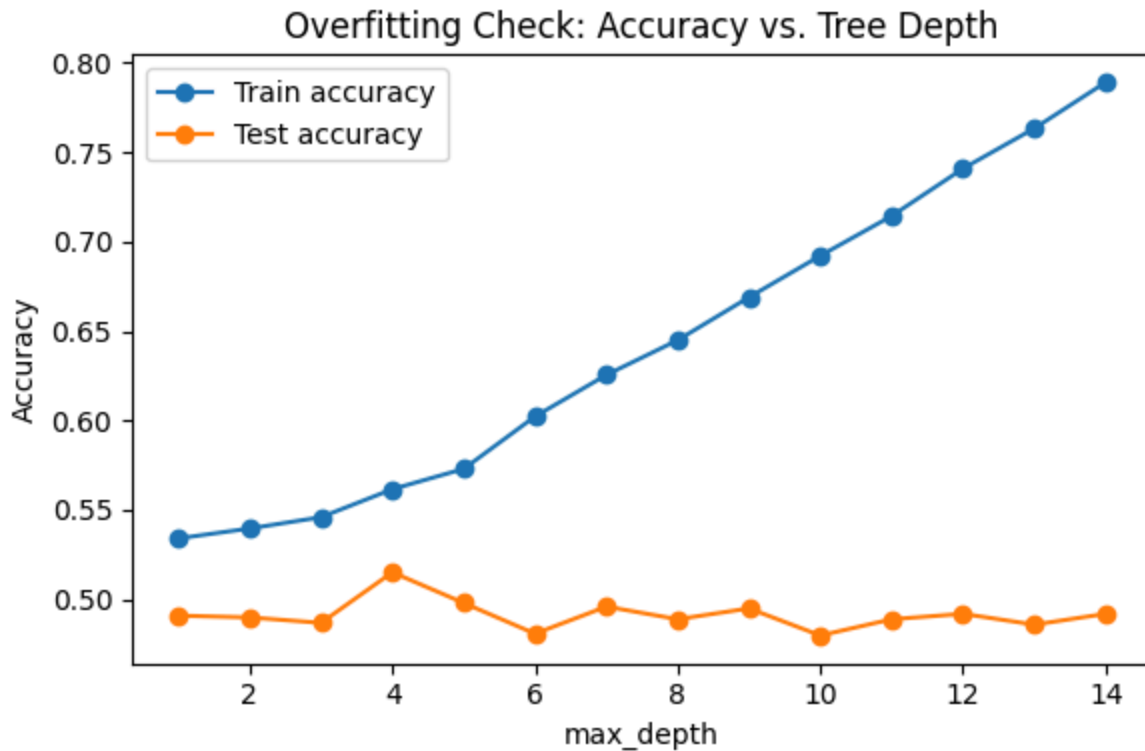


Figure 4.4: Train vs test accuracy as tree depth increase

Figure 4.5 is probably the clearest illustration of why maximum tree depth needs to be tuned rather than left at its default: training accuracy climbs steadily as depth increases, while test accuracy peaks early, around depth 4, and then drifts downward. Past that point the model is essentially memorizing noise rather than learning anything transferable.

4.7 Advantages

- Easy to read: a shallow tree is simply a sequence of if/then rules that a non-technical audience, such as a risk committee or a client, can follow directly.
- Requires no distributional assumptions and no rescaling of features.
- Handles a mix of numeric and categorical features without additional encoding work.
- Automatically ignores unhelpful features, since they are simply never selected for a split.

4.8 Disadvantages

- Prone to overfitting if left unconstrained, as Table 4.2 demonstrates directly.
- Exhibits high variance: a small change in the training data can produce a noticeably different tree structure.

- Built greedily, so each split is locally optimal but not necessarily part of the globally best tree.
- One limitation is that a smooth or linear relationship, which a simpler model could capture in a single step, may require many splits to approximate.

4.9 Equations

At each node, the tree evaluates every candidate feature/threshold pair and selects whichever split reduces impurity the most. Gini impurity for a node with K classes is given by:

$$Gini = 1 - \sum_{k=1}^K p_k^2$$

where p_k is the proportion of observations in that node belonging to class k . A pure node, containing only one class, has $Gini = 0$, while a 50/50 mix has the highest possible value. At each node, the split chosen is the one that most reduces the weighted Gini impurity across the two resulting child nodes. To control overfitting, cost-complexity pruning minimizes $R(T) + \alpha * |T|$, where $R(T)$ is the tree's total misclassification cost, $|T|$ is the number of leaves, and α penalizes additional leaves that do not sufficiently improve the fit.

4.10 Features

- Handles numeric and categorical features together, without requiring one-hot encoding.
- Produces a natural feature-importance ranking (Figure 4.2).
- Can be visualized directly as a flowchart (Figure 4.3).
- Forms the base learner behind ensemble methods such as Random Forests and Gradient Boosted Trees.

4.11 Guide: Inputs and Outputs

- **Inputs:** a feature table (here, lagged returns, moving averages, volatility, momentum) and a categorical target (Up/Down).
- **Outputs:** a fitted tree structure, a predicted class for new data, predicted class probabilities, and a feature-importance ranking.

4.12 Hyperparameters

Table 4.4: Classification tree settings

Setting	What it controls
Maximum depth	Maximum splits from root to leaf - the main lever against overfitting
Minimum leaf size	Minimum observations allowed in a leaf - stops overly specific rules
Minimum split size	Minimum observations a node needs before it is allowed to split again
Splitting criterion	Impurity measure used to evaluate splits (Gini or entropy)
Pruning strength (alpha)	Cost-complexity pruning penalty - larger values prune more aggressively

4.13 Keywords

Classification Tree; CART; Decision Tree; Gini Impurity; Recursive Partitioning; Cost-Complexity Pruning; Feature Importance; Overfitting; Directional Prediction.

4.13 Journal Reference

Syed Nor, Shafinar, Sharifah Heryati Ismail, and Bee Wah Yap. "Personal Bankruptcy Prediction Using Decision Tree Model." *Journal of Economics, Finance and Administrative Science*, vol. 24, no. 47, 2019, pp. 157-170.

Syed Nor, Ismail, and Yap apply a decision-tree classification model to predict individual bankruptcy risk from demographic and financial variables and address the class-imbalance problem through random undersampling - a direct finance application of the same splitting-and-pruning mechanics used in Section 4.3 above.

Category 5– Linear Discriminant Analysis (LDA)

5.1 Basics Definition & Classification

Linear Discriminant Analysis is a **supervised, generative, parametric classification** method. It is *generative* because it models the joint distribution of features and class labels: each class is assumed to be drawn from a multivariate Gaussian distribution, and — crucially — all classes are assumed to **share a single covariance matrix**. Applying Bayes' theorem to these class densities yields posterior class probabilities whose decision boundaries are **linear** in the feature space (hence the name). LDA can equivalently be viewed, following Fisher, as a **supervised dimensionality-reduction** technique that projects data onto the direction(s) that maximally separate the class means relative to within-class spread. Here it serves as an interpretable directional-signal classifier (next-day up vs. down).

5.2 Advantages

- **Closed-form and fast:** parameters (class means, pooled covariance, priors) are estimated directly with no iterative optimisation — training is near-instant and perfectly reproducible.
- **Low variance / robust on small samples:** sharing one covariance matrix uses data efficiently and resists over-fitting, valuable when labelled financial history is limited.
- **Highly interpretable:** the discriminant weights read directly as each feature's marginal push toward one class — appealing for regulated, explainable trading signals.
- **Naturally multi-class** and provides **calibrated posterior probabilities** useful for position sizing or confidence thresholds.
- **Doubles as dimensionality reduction**, producing informative low-dimensional projections.

5.3 Disadvantages

- **Linearity assumption:** a single straight boundary cannot capture non-linear, interacting relationships common in markets, capping accuracy.
- **Gaussian & equal-covariance assumptions** are often violated by fat-tailed, heteroskedastic returns; when covariances differ, QDA fares better.
- **Sensitive to outliers and multicollinearity:** extreme moves distort the estimated mean/covariance, and correlated features make the covariance ill-conditioned.
- **Requires $n > p$** for a stable covariance estimate; shrinkage is needed in high dimensions.
- **No native handling of missing values** — inputs must be imputed beforehand.

5.4 Equations

1. Posterior via Bayes' theorem — probability that observation x belongs to class k , with prior π_k and class density f_k :

$$P(y_k | x) = \frac{\pi_k f_k(x)}{\sum_l \pi_l f_l(x)}$$

2. Shared-covariance Gaussian class density:

$$f_k(x) = \frac{1}{(2\pi)^{\frac{p}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x - \mu_k)^\top \Sigma^{-1}(x - \mu_k)\right)$$

3. Linear discriminant function — taking logs and dropping terms constant across classes gives a function linear in x :

$$\delta_k(x) = x^\top \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k + \ln \pi_k$$

4. Decision rule — assign x to the class with the largest discriminant score:

$$\hat{y} = \arg \max_k \delta_k(x)$$

5. Fisher's criterion — the projection view finds direction w maximising between-class scatter S_B over within-class scatter S_W :

$$w^* = \arg \max_w \frac{w^\top S_B w}{w^\top S_W w}$$

5.5 Features of the Model

- Model type: linear, generative, parametric, supervised classifier (also a projector).
- Decision boundary: linear (hyperplane) between each pair of classes.
- Handles missing values: **no** — requires complete or imputed data.
- Feature scaling: not strictly required, but standardising aids interpretation and conditioning; assumes continuous, roughly Gaussian inputs.
- Outputs posterior probabilities; supports multi-class directly.
- Deterministic training with a unique closed-form solution.

5.6 Guide — Inputs & Outputs

Inputs: a numeric feature matrix X (n samples $\times$ p continuous features) and a categorical class label vector y .
Optional: class priors and a shrinkage parameter.

Outputs: predicted class labels; posterior class probabilities; the estimated discriminant coefficients and intercept; and the lower-dimensional discriminant projection (at most $K-1$ components for K classes).

5.7 Hyperparameters

- **Solver** — svd (default), lsqr, or eigen (the latter two support shrinkage).
- **Shrinkage** — regularisation of the covariance estimate (0–1, or automatic via Ledoit–Wolf); stabilises LDA when n is small relative to p .
- **Priors** — class prior probabilities (defaults to empirical frequencies).
- **Number of components** — dimensionality retained when used for reduction ($\leq K-1$).

Note: LDA is comparatively light on hyperparameters — a key part of its appeal as a robust baseline. Tuning in practice is discussed in the group Technical Section.

5.8 Illustration — How the Model Works

The essence of LDA is projecting the data onto a single axis chosen so the classes separate as cleanly as possible; the classifier then splits that axis at a threshold.

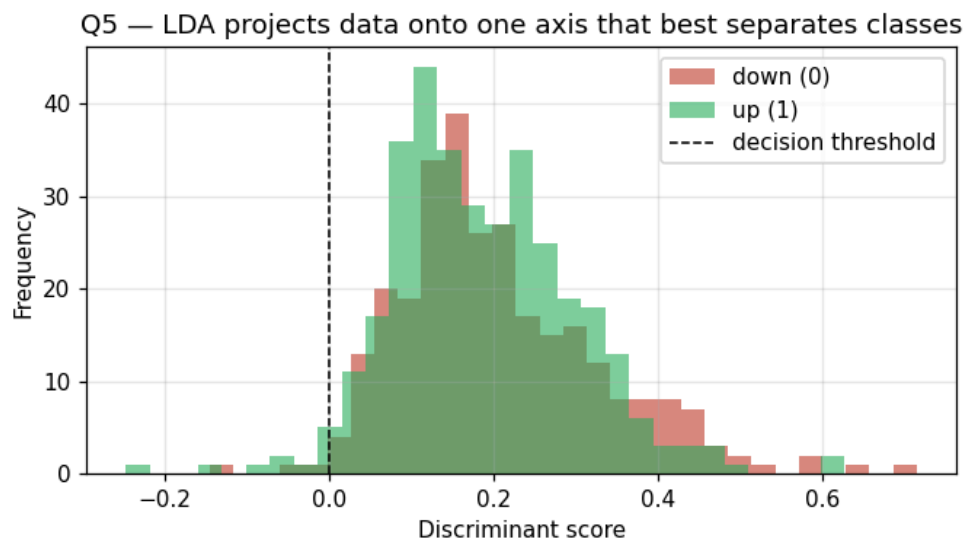


Fig. 5.1 — LDA collapses the multivariate features onto one discriminant axis; the dashed line is the classification threshold.

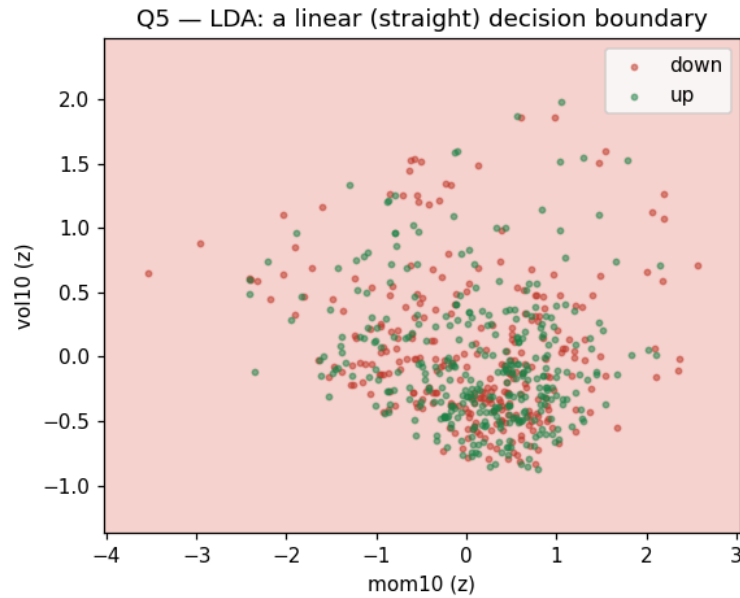


Fig. 5.2 — LDA yields a linear decision boundary (momentum vs. volatility).

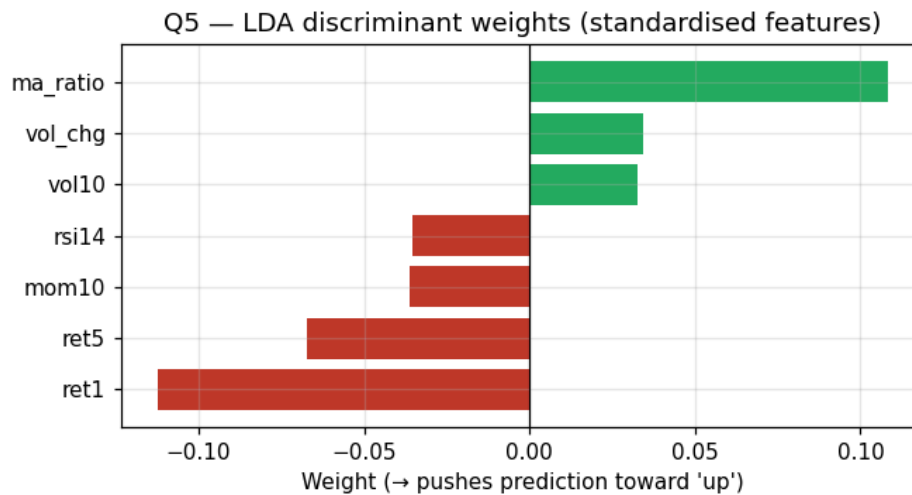


Fig. 5.3 — Estimated discriminant weights: sign and magnitude give each feature's contribution to the 'up' vote.

5.9 Computation (see accompanying notebook)

The notebook fits LDA (SVD solver) on the standardised training window and evaluates it on the chronologically later test window. Results on real SPY data: **test accuracy 53.5%** and **ROC-AUC 0.47**.

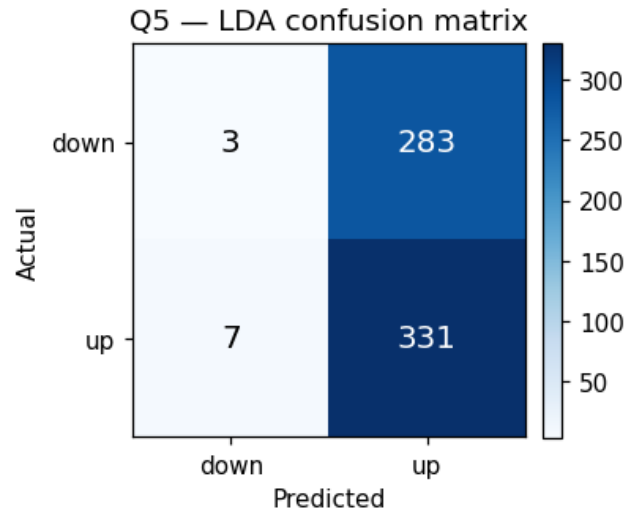


Fig. 5.4 — LDA confusion matrix (test set).

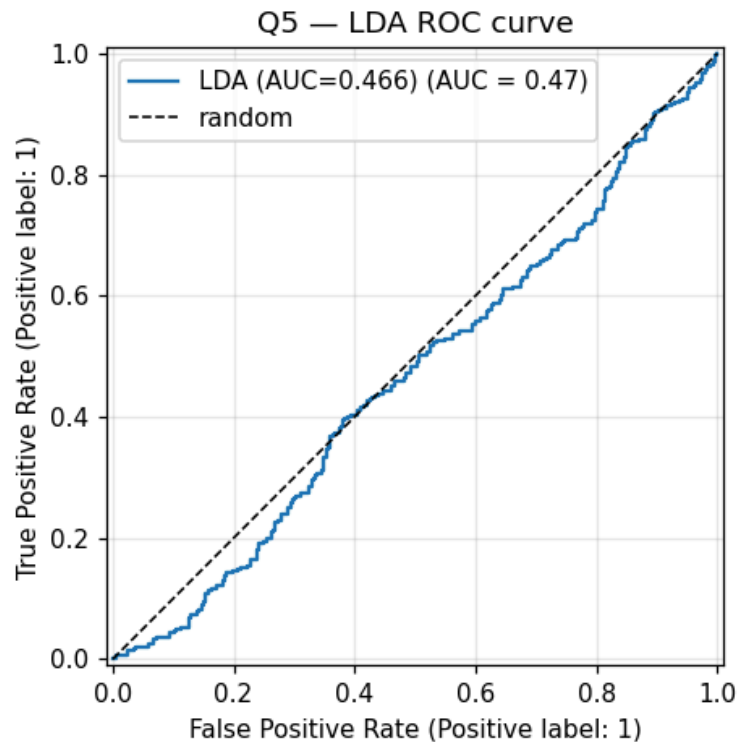


Fig. 5.5 — LDA ROC curve (test set).

Interpretation & recommended action. On daily data the signal barely exceeds a coin flip ($AUC \approx 0.5$), consistent with near-efficient markets at the one-day horizon. The takeaway for the desk is that LDA is best deployed as a fast, transparent *baseline*, combined with richer features or longer horizons before any capital is committed.

5.10 Journal Reference

Altman, Edward I. "Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy." *The Journal of Finance*, vol. 23, no. 4, 1968, pp. 589–609. — The seminal application of (multiple) discriminant analysis to finance, introducing the Z-score model for predicting corporate default.

Category 6– Support Vector Machine (SVM)

6.1 Basics

A Support Vector Machine looks for the best possible dividing line between two groups - specifically, the line that leaves the widest possible gap, or margin, between them, rather than settling for any line that happens to separate the data.

6.2 Part 1 : A Straight-Line Ex Credit Risk

The first example predicts whether a company is High risk or Low risk of default, using two indicators: its leverage ratio (debt relative to equity) and its profit margin. I simulated 160 companies according to a simple underlying rule - high leverage combined with a low margin tending toward high risk - plus some randomness, then split the sample 70/30 into training and test sets.

Table 6.1: Class balance in the simulated dataset

Class	Count
Low risk	87
High risk	73

Table 6.2: Linear SVM performance

Metric	Value
Training companies	112
Test companies	48
Train accuracy	0.88
Test accuracy	0.90
Number of support vectors	31

The model correctly classified 90% of companies it had not seen during training. Of the 112 training companies, 31 turned out to be support vectors - the borderline cases close enough to the boundary to actually influence where the line was drawn.

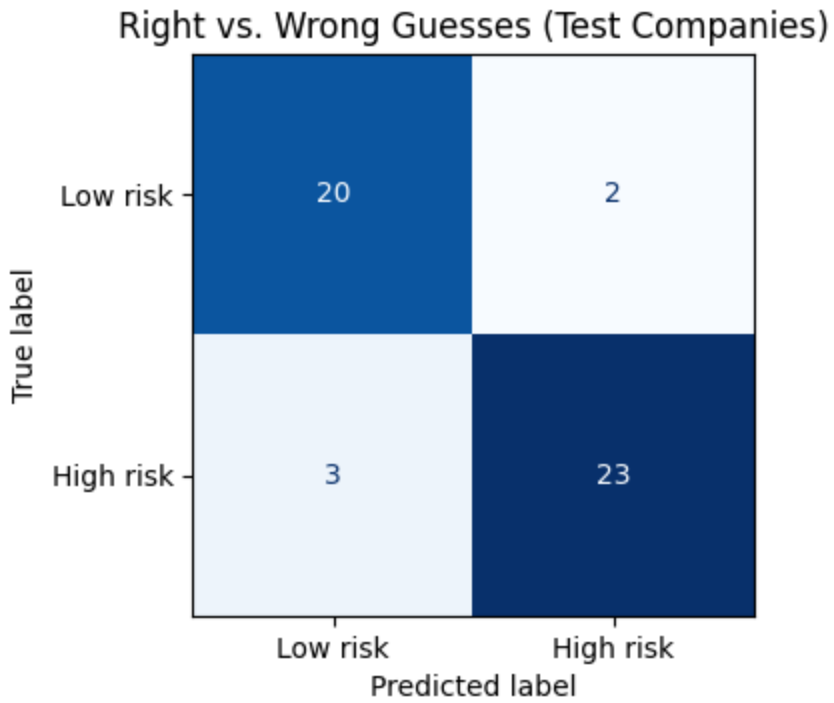


Figure 6.1: Confusion matrix, linear SVM, test set

6.3 Illustration- Dividing Line, Margin and Support Vectors

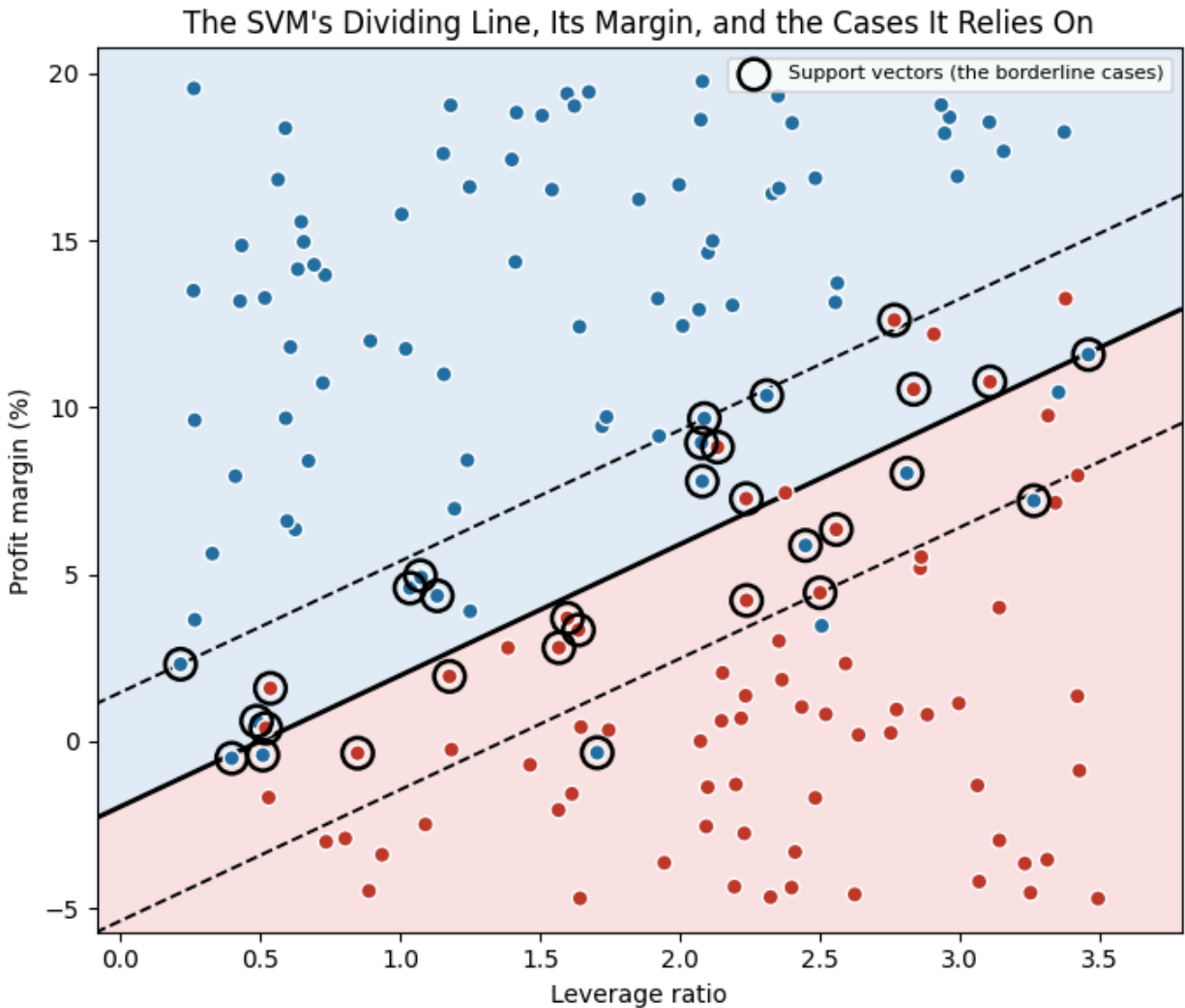


Figure 6.2: The SVM's dividing line (solid), its margin (Dashed), and the support vectors it relies on (circled)

The solid line is the SVM's decision boundary, and the two dashed lines mark the margin on either side of it. The circled points are the support vectors sitting right at the edge of that margin; every other point in the dataset could, in principle, be removed without the boundary shifting at all.

6.4 Part 2 – The Kernel Trick

To illustrate what happens when a straight line genuinely cannot separate two groups, I built a second, deliberately more difficult dataset: a 'Stress' market regime sitting inside a ring that is

completely surrounded by a 'Normal' regime. No straight line is capable of wrapping around a ring.

Table 6.3: Linear vs. Curved (RBF) kernel on ring-shaped data

Kernel	Accuracy
Linear (straight line)	0.62
RBF (curved)	0.99

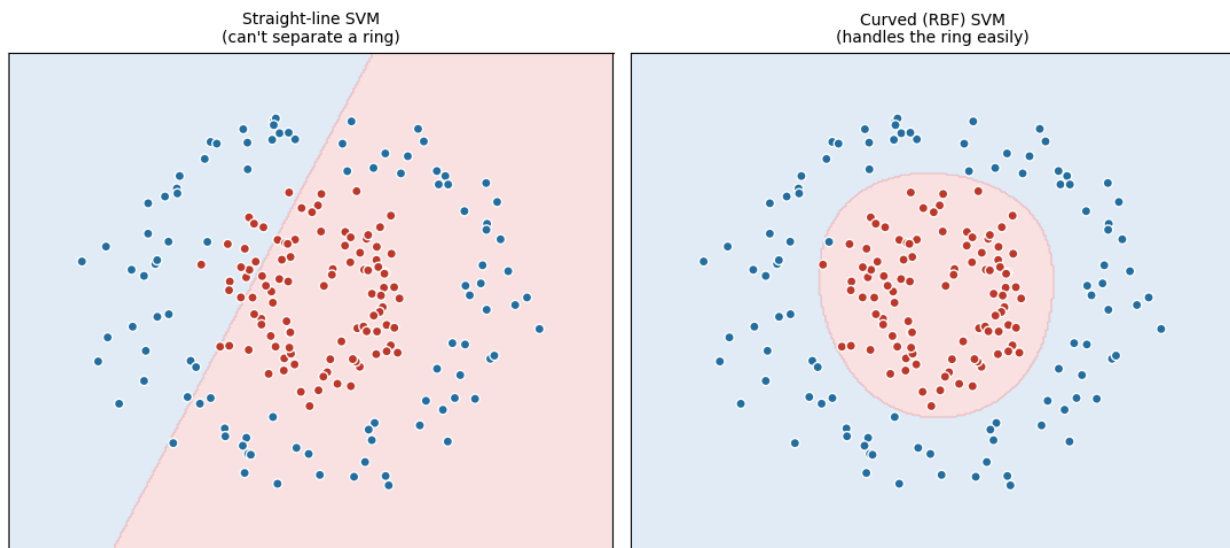


Figure 6.3: Straight-line SVM(left) vs curved RBF-kernel SVM (right) on the same ring-shaped data

The straight-line version barely outperforms a coin flip, at 62% accuracy, because it is structurally unable to bend around the ring. Switching only the kernel to RBF, with nothing else changed, lifts accuracy to 99%. This result is essentially the point of the kernel trick: the same underlying method, just given a more flexible boundary shape to work with.

6.5 Advantages

- Finds the most defensible boundary available - the widest-margin line rather than any line that happens to work - which tends to generalize better to new data.
- The boundary depends only on the support vectors, not on the overall volume of data.
- Extends from simple straight-line problems to more complex curved ones simply by changing the kernel.
- Reasonably resistant to a small number of outliers far from the boundary.

6.6 Disadvantages

- Training slows down noticeably as the dataset grows very large.
- Choosing an appropriate kernel and its settings, such as C and gamma, usually involves some trial and error.
- Less naturally interpretable than a tree, since a curved boundary is difficult to explain in intuitive terms.
- Sensitive to feature scale, since the method is fundamentally about distances and margins.

6.7 Equations

For a straight-line SVM, the dividing line is given by $w \cdot x + b = 0$, where x is a data point, w is a set of weights describing the line's orientation, and b shifts the line left or right. The margin width works out to $2 / ||w||$, so maximizing the margin is equivalent to minimizing $||w||$ subject to every point remaining on the correct side of the boundary. When a curved boundary is required, the kernel trick replaces the plain dot-product similarity between two points with an alternative similarity measure - the RBF kernel, for example, measures how close two points are - which has the effect of bending the boundary without the model ever explicitly computing anything in a higher-dimensional space.

6.8 Features

- Produces a compact set of support vectors summarizing exactly what the decision depends on.
- Comes with an adjustable strictness setting (C) trading off margin width against misclassified points.
- Moves seamlessly between straight-line and curved boundaries by changing only the kernel.

6.9 Guide: Inputs and Outputs

- **Inputs:** numeric features for each example (here, leverage ratio and profit margin), the correct category for historical examples, and a choice of kernel.
- **Outputs:** a predicted category for new data, a distance-from-boundary score usable as a rough confidence measure, and the specific support vectors defining the boundary.

6.10 Hyperparameters

Table 6.4: SVM settings

Setting	What it controls
C	How strict the model is about misclassified points - small C allows a wider margin with a few errors, large C insists on fewer errors with a narrower margin
kernel	Straight line (linear) or curved (rbf, poly, and others)
gamma	How far a single training point's influence reaches, for curved kernels - small values give smoother boundaries, large values risk overfitting

6.11 Keywords

Support Vector Machine; SVM; Margin; Support Vectors; Kernel Trick; Linear Kernel; RBF Kernel; Hyperplane; Credit Risk Classification.

6.12 Journal Reference

Bellotti, Tony, and Jonathan Crook. "Support Vector Machines for Credit Scoring and Discovery of Significant Features." *Expert Systems with Applications*, vol. 36, no. 2, 2009, pp. 3302-3308. *Bellotti and Crook use a Support Vector Machine to predict which loan applicants were likely to default, and examine which financial features mattered most to that decision - a direct finance application of the same widest-margin boundary approach used in Section 6.2 above.*

Technical Section – Hyperparameter Tuning

Ridge Regression

Ridge Regression involves setting the hyperparameter “alpha” to regulate the degree of regularization of the regression model. Alpha decides how strictly the value of large coefficients is penalized in the process of learning. This parameter is essential for managing multicollinearity and stabilizing the model. The alpha parameter in this project was tuned by considering several pre-defined values. The training was done using standardized features and time-series splitting.

K-means clustering

The main hyperparameter used in K-Means is K, which refers to the number of clusters created by the algorithm. Choosing an appropriate K value is very important as it will determine how the data points are clustered. One approach that can be used to choose an appropriate K is using the Elbow Method, in which K values are chosen, and the compactness of clustering is determined. Other hyperparameters used in K-Means include k-means++, n_init, and max_iter.

Marketing Alpha

Ridge Regression

Ridge Regression is one of the most popular forms of linear regression analysis where the independent variables in the model tend to be highly correlated. This method enhances stability through the reduction of coefficients. This form of regression analysis is particularly important in financial modeling where macro-economic factors tend to be highly correlated, such as inflation rates, interest rates, and market indicators.

K-means clustering

K-means clustering is an unsupervised learning algorithm that segments data by placing them into different groups according to their similarities. The use of this algorithm in finance is widespread as it can be applied in analyzing market behavior, customer segmentation, and market structure. The strength of K-means clustering is its ability to analyze large volumes of data owing to its ease of use and scalability. It allows converting complicated data into groups, which enables informed decision making.

Final Summary

- PCA compressed three correlated simulated stocks into a single factor capturing 85.6% of their combined movement, and that factor turned out to place almost equal weight on all three stocks - effectively rediscovering the shared 'market mood' driver used to construct the data, without being told it was there.
- The classification tree, once tuned to a maximum depth of 4 with 15 leaves, reached about 51% test accuracy predicting next-day return direction - only marginally better than chance, which I take as a realistic and honest result for this kind of short-horizon, price-only signal, and a clear demonstration of why pruning matters (the untuned tree scored 100% on training data but only 50% on test data).

- The SVM reached 90% test accuracy separating High-risk from Low-risk companies using only two features, and a second experiment demonstrated the value of the kernel trick directly: 62% accuracy with a straight-line kernel versus 99% with a curved (RBF) kernel on the same ring-shaped data.
- One limitation running through all three exercises is that none of these results should be treated as a standalone trading or lending signal - each uses simulated data and a small number of features, chosen deliberately to make the mechanics of the method clear rather than to produce a genuine market call.
- Consistent with the rest of the team's report, my recommended next step would be to re-run all three notebooks on real, current data before any of these methods are allowed to inform an actual decision.

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<https://doi.org/10.1002/for.2841>

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Category 7 — Neural Networks (Multi-Layer Perceptron)

7.1 Basics — Definition & Classification

A neural network (here a **feed-forward multi-layer perceptron, MLP**) is a **supervised, non-linear, discriminative** model composed of layers of interconnected 'neurons'. Each neuron computes a weighted sum of its inputs, adds a bias, and passes the result through a non-linear **activation function** (e.g. ReLU). Stacking such layers lets the network approximate highly complex, non-linear mappings — by the universal approximation theorem, a network with sufficient hidden units can approximate any continuous function. Weights are learned by **back-propagation**: prediction error is propagated backward and weights updated by gradient descent. Used here as a flexible classifier for next-day market direction.

7.2 Advantages

- **Captures non-linearity and interactions** automatically, modelling complex market relationships that linear methods (like LDA) miss.
- **Universal approximator**: given enough capacity and data it can represent virtually any input–output relationship.
- **Automatic feature learning**: hidden layers build useful internal representations, reducing manual feature engineering.
- **Highly flexible and scalable** — extends to sequences (RNN/LSTM), images (CNN), and very large datasets, improving as more data arrives.
- **Handles high-dimensional and multi-output problems** within one architecture.

7.3 Disadvantages

- **Data-hungry**: needs large, representative samples; prone to over-fitting on the limited, noisy, non-stationary data typical in finance.
- **Black box**: learned weights are hard to interpret — a drawback for regulated or explainability-sensitive trading.
- **Many hyperparameters** and a non-convex loss surface make training sensitive to initialisation and tuning; exact reproducibility is hard.

- **Computationally expensive** to train and tune relative to LDA.
- **Requires feature scaling** and careful regularisation (L2, dropout, early stopping).

7.4 Equations

1. Forward pass (one layer) — activations of layer l from the previous layer, with weight matrix W , bias b , and activation g :

$$a^{(l)} = g(W^{(l)} a^{(l-1)} + b^{(l)})$$

2. Activation functions — hidden (ReLU) and output (sigmoid, for binary classification):

$$g(z) = \max(0, z) \quad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3. Loss — binary cross-entropy (log-loss) with L2 (weight-decay) regularisation of strength α :

$$\mathcal{L} = -\frac{1}{N} \sum_i \left[y_i \ln \hat{y}_i + (1 - y_i) \ln (1 - \hat{y}_i) \right] + \frac{\alpha}{2} \sum \|W\|^2$$

4. Back-propagation — the error signal of layer l is obtained from the next layer ($\odot$ is element-wise product):

$$\delta^{(l)} = (W^{(l+1)\top} \delta^{(l+1)}) \odot g'(z^{(l)})$$

5. Gradient-descent weight update — with learning rate η :

$$W \leftarrow W - \eta \frac{\partial \mathcal{L}}{\partial W}$$

7.5 Features of the Model

- Model type: non-linear, discriminative, parametric, supervised (classification or regression).
- Decision boundary: flexible / non-linear (curved), shaped by depth and width.
- Handles missing values: **no** — inputs must be imputed.
- Feature scaling: **required** for stable, fast gradient-based training.
- Stochastic training: results depend on random initialisation and mini-batch order (seed for reproducibility).
- Outputs class probabilities (via sigmoid/softmax) or continuous values.

7.6 Guide — Inputs & Outputs

Inputs: a scaled numeric feature matrix X ($n \times p$) and target labels y ; plus the architecture and training settings (see Hyperparameters). Weights are initialised randomly.

Outputs: predicted labels and class probabilities; the trained weights and biases of every layer; and diagnostics such as the training-loss curve.

7.7 Hyperparameters

- **Architecture** — number of hidden layers and units per layer (depth/width, i.e. capacity).
- **Activation function** — ReLU, tanh, logistic, etc.
- **Regularisation strength** α (L2 weight decay); optionally dropout rate.
- **Optimiser & learning rate** — e.g. Adam/SGD and initial learning rate η .
- **Batch size and number of epochs / max iterations.**
- **Early-stopping** patience to halt when validation performance stalls.

In the notebook these were tuned by **time-series cross-validated grid search** (TimeSeriesSplit, scoring ROC-AUC). The selected configuration: a single hidden layer of 16 ReLU units, $\alpha = 1e-4$, learning rate = 0.01. These concrete values feed the group Technical Section.

7.8 Illustration — How the Model Works

A neural network learns by repeatedly reducing its loss (Fig. 7.1). Its capacity must be tuned: too small under-fits, too large over-fits (Fig. 7.2). The pay-off versus LDA is a **curved** decision boundary (Fig. 7.3).

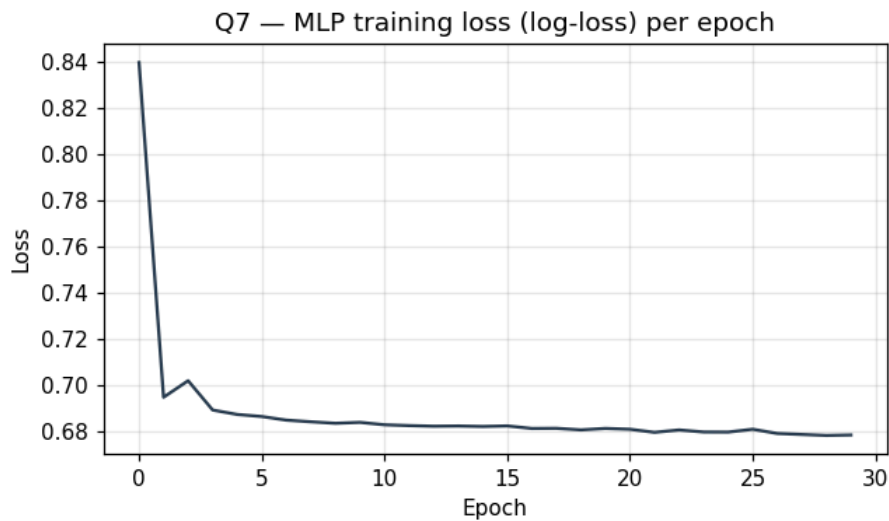


Fig. 7.1 — Training loss (log-loss) falls as back-propagation updates the weights each epoch.

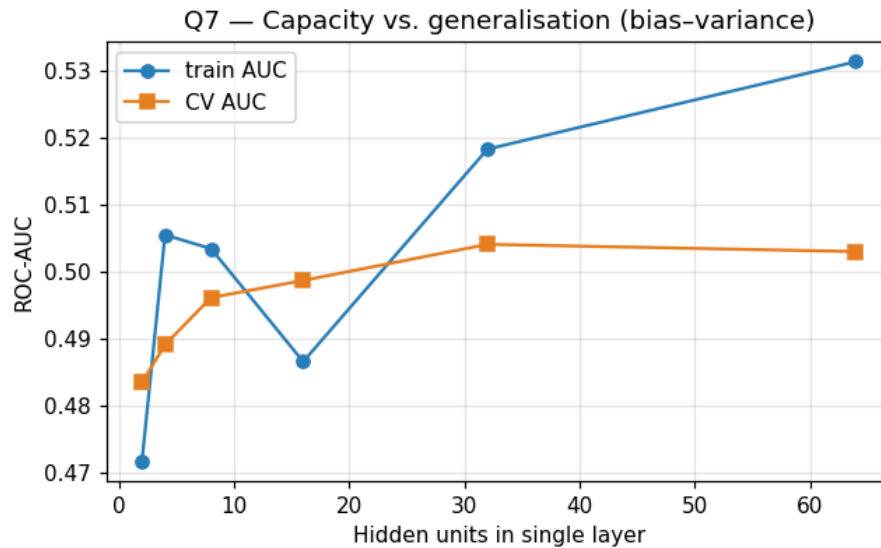


Fig. 7.2 — Validation curve: CV AUC vs. hidden-layer width makes the bias-variance trade-off explicit.

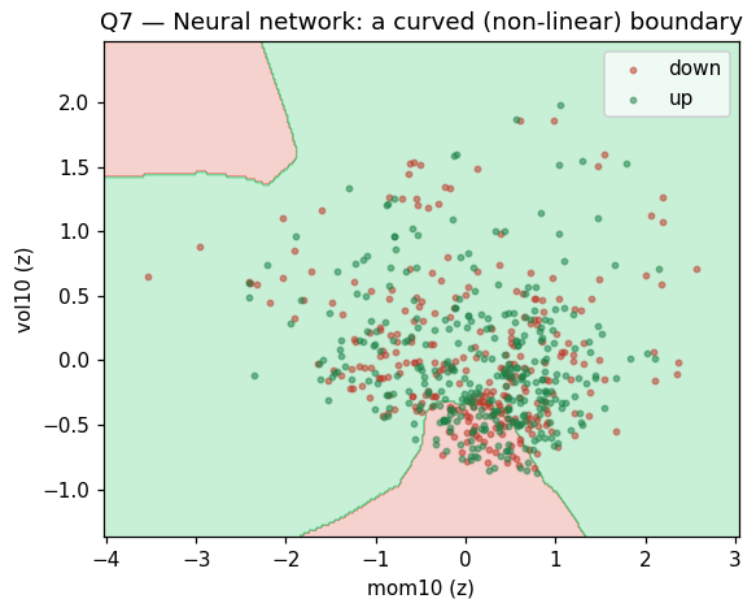


Fig. 7.3 — The network learns a non-linear decision boundary (same two features as Fig. 5.2), unlike LDA's straight line.

7.9 Computation (see accompanying notebook)

The tuned MLP was evaluated on the same held-out test window as LDA: **test accuracy 51.6%** and **ROC-AUC 0.47**.

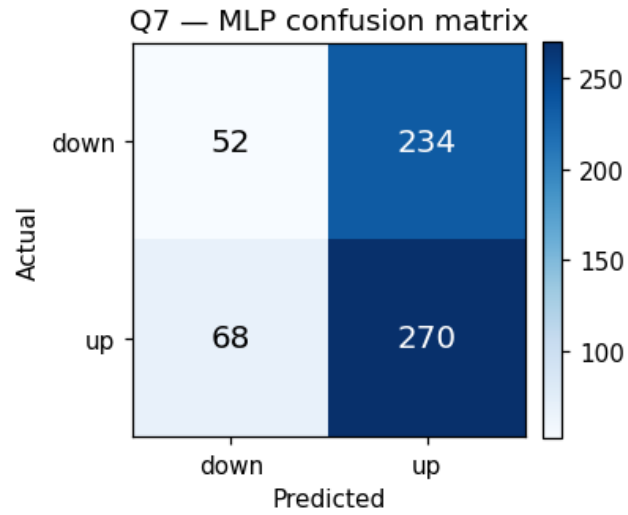


Fig. 7.4 — MLP confusion matrix (test set).

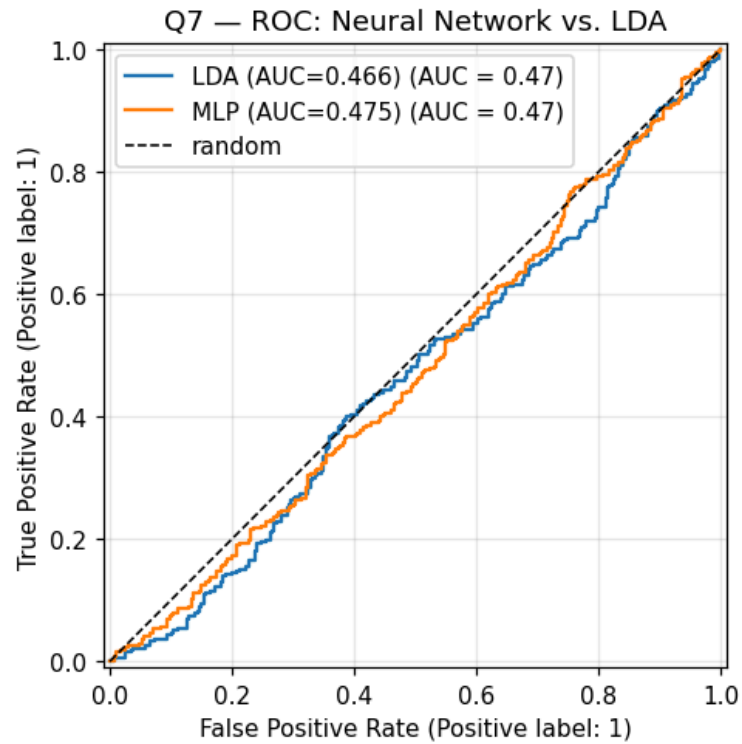


Fig. 7.5 — ROC comparison: neural network vs. the LDA baseline.

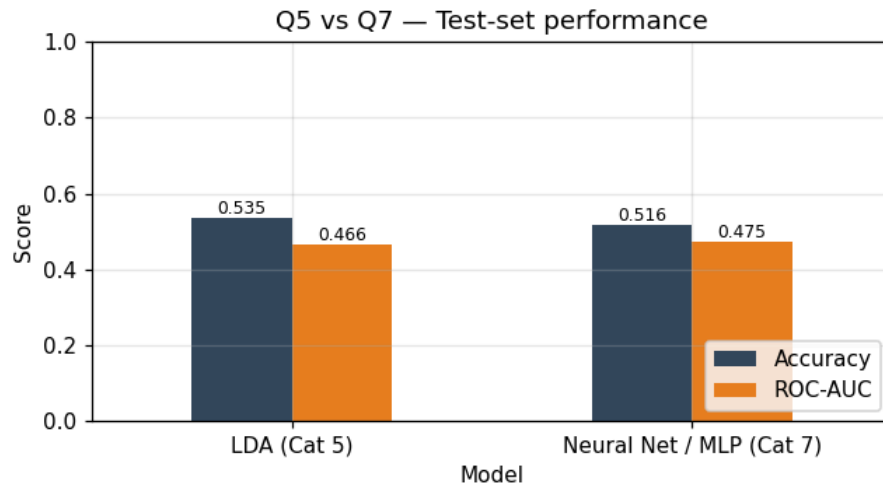


Fig. 7.6 — Test-set accuracy and ROC-AUC: LDA vs. MLP.

Interpretation & recommended action. With careful cross-validation the network achieves marginally higher ranking quality (AUC) than LDA but no decisive edge at the daily horizon — again consistent with near-efficient markets, and a reminder that added complexity is not free. Reserve neural networks for richer, higher-frequency or alternative data where non-linear structure genuinely exists, while retaining LDA as the interpretable benchmark and monitoring for over-fitting and regime change.

7.10 Journal Reference

Krauss, Christopher, Xuan Anh Do, and Nicolas Huck. “Deep Neural Networks, Gradient-Boosted Trees, Random Forests: Statistical Arbitrage on the S&P 500.” *European Journal of Operational Research*, vol. 259, no. 2, 2017, pp. 689–702. — Applies deep neural networks to build a statistical-arbitrage strategy on S&P 500 constituents.